Exp-a

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End-to-End communication at transport layer :-

(A) Echo client-server using TCP

Aim: To implement echo client server using TCP packets.

Procedure:-

- Create socket using python socket web programming
- Bind server socket to IP address and ports, then listen.
- Connect client to server, send message, receive echo, and close socket.

Code:-

(i) TCP server (tcp_server.py)

```
import socket
s= socket.socket(socket.AF_INET, Socket.SOCK_STREAM)
s.bind(("127.0.0.1", 5000))
s.listen()
print("server waiting...")
conn, addr= s.accept()
print("client connected")
while True:
    data = conn.recv(1024).decode()
    if not data:
        break
    print("client:, data
```

conn.send (data.encode)

conn.close()

s.close()

(ii) TCP client (tcp-client.py)

```
import socked
c=socket.socked(socket.AF_INET,
Socked.SOCK_STREAM)
c.connect (("127.0.0.1,5000))
msg = input ("Enter msg:")
c.send (msg.encode())
reply = c.recv(1024).decode
print ("server, reply)
c.close()
```

Output:-

Terminal Server	Terminal client
server waiting	Says HI
client cond	server: HI
Client: HI	TCP Echo is working.

Result:-

The TCP packets using echo client server socket program is executed successfully.

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(B) Chat client server using TCP :-

Aim :- To implement chat client server using TCP packet.

(i) Chat server (chat_server.py)

```
import socket
s = socket.socket(socket.AF_INET, socket.SOCK_
    STREAM)
s.bind(("127.0.0.1", 6000))
s.listen(1)
print("Waiting")
conn, addr = s.accept()
print("Client connected")

while True:
    msg = conn.recv(1024).decode()
    if msg.lower() == "bye":
        print("client ended chat")
        break
    print("client:", msg)
    reply = input("Server:")
    conn.send(reply.encode())
    if reply.lower() == "bye":
        break
conn.close()
s.close()
```

(ii) Chat client (chat_client.py)

```
import socket
c=socket.socket(socket.AF_INET,
socket.SOCK_STREAM)
c.connect(("127.0.0.1", 6000))
print("connected to server")
while True:-
    msg = input("client").
    c.send(msg.encode())
    if msg.lower() == "bye":
        break
c.close()
```

Output:-

Server	Client
wait for client	connects Faster
client connected	client: Hi
client: HI	server! Hello
sevi Hello	server. How are you
Server: Fine	

Result:-

The chat programming using packets using sockets by TCP client & server was implemented successfully.

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